End-to-End Cleaning, Outlier Detection, and CSV Export:

In [26]:
```
import pandas as pd

df = pd.read_csv("/content/bookbazaar_orders_2026.csv")
print("Original shape:", df.shape)
print("Missing values:", df.isnull().sum().sum())
```

```
Original shape: (86, 19)
Missing values: 6
```

In [27]:
```
df = df.drop_duplicates(subset="Order_ID", keep="first").copy()
df["City"] = df["City"].str.strip().str.title()
df["Delivery_Status"] = df["Delivery_Status"].str.strip().str.title()
df["Customer_Name"] = df["Customer_Name"].fillna("Unknown")
```

In [28]:
```
df["Unit_Price"] = pd.to_numeric(df["Unit_Price"], errors="coerce")
df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")
df["Ship_Date"] = pd.to_datetime(df["Ship_Date"], errors="coerce")
```

In [29]:
```
print("Missing or invalid values still requiring review:")
print(df.isnull().sum()[df.isnull().sum() > 0])
```

```
Missing or invalid values still requiring review:
Unit_Price          1
Discount_Percent    1
Shipping_Cost       1
Order_Date          1
Ship_Date           2
Rating              1
dtype: int64
```

In [30]:
```
Q1 = df["Unit_Price"].quantile(0.25)
Q3 = df["Unit_Price"].quantile(0.75)
IQR = Q3 - Q1
outliers = df[(df["Unit_Price"] < Q1 - 1.5 * IQR) | (df["Unit_Price"] > Q3 + 1.5
print("Price outliers:", len(outliers))
```

```
Price outliers: 0
```

In [31]:
```
df.to_csv("/content/bookbazaar_orders_2026_cleaned.csv", index=False)
print("Cleaned shape:", df.shape)
print("Saved: bookbazaar_orders_2026_cleaned.csv")
```

```
Cleaned shape: (84, 19)
Saved: bookbazaar_orders_2026_cleaned.csv
```